MXB109 Exam Cheat Sheet

Everything you need in the exam: **which method**, **the rules**, **the formulas**, and **the sentence to write**.

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0. The routine for every question

1. **Diagnose:** "This is a _____ problem because _____." (1 line, usually worth marks)
2. **Run the method:** show every table, the entering/leaving variable, and the row operations.
3. **Answer in words, with units:** "Make 20 chairs and 10 shelves for $13,000 profit. Timber and finishing are fully used; 10 labour days are spare."

Which method?

The question has...	Method	Note
supplies, demands, a cost grid	Transportation	7
two opponents, a payoff table	Game theory	8
nodes and roads/pipes, "quickest route"	Shortest path	9
nodes and roads, "connect everything, least cable"	Minimum spanning tree	9
an LP with **2 variables**	Graphical	2
an LP, all $\leq$, RHS ≥ 0 (origin works)	Simplex	3
an LP with $\geq$ or $=$ (origin fails)	Big M, then simplex	4
"one more unit worth", "binding", "range"	Shadow price	5
"not produced", "how much must profit rise", "new product"	Reduced cost	5
"dual", "bound", "certify optimal", "value of resources"	Duality	6

1. Formulation

Three parts, always:

1. **Decision variables**, with units: "x_1 = kg of brand A per week". Never call a variable z.
2. **Objective:** max profit or min cost, with units ("z in $100s").
3. **Constraints**, one per limit/requirement, each labelled, plus $x_j \geq 0$.

Words	Symbol
"at most", "no more than", "up to", "available", "capacity"	$\leq$
"at least", "no less than", "minimum", "requirement"	$\geq$
"exactly", "all of", "must total"	$=$
"A is at least 30% of the total"	$x_A \geq 0.3(x_A + x_B + x_C) \Rightarrow \mathbf{0.7x_A - 0.3x_B - 0.3x_C \geq 0}$
"at least twice as many A as B"	$x_A \geq 2x_B \Rightarrow \mathbf{x_A - 2x_B \geq 0}$

Reading it: always move every variable to the left-hand side so the RHS is a plain number.

Check every coefficient against the table in the question (your Exam 2 typo: 0.01 instead of 0.10 cost you two parts).

2. Standard form: slack, excess, artificial

Constraint	Add	Name	Formula	Meaning
$\leq$	$+\mathbf{s}$	slack	$\mathbf{s} = \text{RHS} - \text{LHS}$	how much is **left over / unused**
$\geq$	$-\mathbf{e}$ $(+a)$	excess (surplus)	$\mathbf{e} = \text{LHS} - \text{RHS}$	how much you went **over** the requirement
$=$	$+\mathbf{a}$	artificial only	none (no slack)	nothing real; must be 0 at the end

Reading it: slack and excess are always ≥ 0. A negative one means the plan **breaks** that constraint (infeasible).

Value	Word	Sentence for the exam
s or $e = 0$	**binding**	"All 18 kiln hours are used up; kiln time is what limits the plan."
s or $e > 0$	**non-binding**	"4 litres of glaze are left over; glaze isn't limiting anything."

Explain without the words "slack" or "binding" (asked in Exam 1): "There are 4 kiln hours left over, so we could make more mugs and earn more. So this plan can't be the best."

Before you start:

- **Negative RHS:** multiply the row by -1 (flips $\leq \leftrightarrow \geq$).
- **Min problem:** $\max z' = -z$. At the end, $\mathbf{z} = -\mathbf{z}'$.
- **How many artificials?** Plug the origin into every row. Every row that **fails** (a $\geq$ with positive RHS, or an $=$) gets one. A $\geq$ with RHS 0 passes, so it needs none. Say which rows and why.

Basic solutions: n variables (after adding slacks) and m equations $\Rightarrow$ $n - m$ **variables are 0** at a basic solution; there are at most $\binom{n}{m}$ of them. Set the chosen ones to 0 and solve for the rest; feasible only if all are ≥ 0.

3. Graphical method

1. Draw each constraint line (find both intercepts), shade the allowed side, label the lines.
2. List **every corner** and its z value in a table. (Missing z values = lost marks.)
3. Pick the best z. Or slide the isoprofit line: max $\rightarrow$ furthest out, min $\rightarrow$ closest in.

Slope of $ax + by = c$ is $-a/b$.

Objective coefficient range (how far a price can move before the plan changes):

1. Find the two **binding** lines at the optimal corner.
2. Work out their slopes $-a/b$.
3. Require the profit slope $-c_1/c_2$ to sit **between** them; solve for the changing price. (Multiplying by a negative flips both inequalities.)

Picture	Meaning
objective parallel to a binding edge	**alternative optima**: every point on that edge (give the segment, e.g. $2x_1 + x_2 = 16$, $3 \leq x_1 \leq 6$)
region open in the improving direction	**unbounded**
no overlapping region	**infeasible**
min problem, open region, positive costs	still has an optimum: cost only grows as you move further out, so the minimum is at a corner

Shadow price graphically: RHS $+1$, re-solve, **new** z $-$ **old** z.

4. Simplex

Step	Rule
Start	slacks are basic; you're at the origin
Enter	most negative number in the z row (variable columns only)
Leave	ratio RHS $\div$ entering-column entry, **positive entries only**, smallest wins
Pivot	divide the pivot row by the pivot; then **new row = old row − (its entry in pivot column) × new pivot row**
Stop	no negatives in the z row (max)

Write the row ops every pivot: $R_2' = R_2/3,\ R_1' = R_1 - R_2',\ R_z' = R_z + 4R_2'$.

Checks after every pivot: entering column = 1 with 0s elsewhere (z row too); all RHS ≥ 0; z never gets worse.

Reading any table

Look at	It tells you
Basis column + RHS	the basic variables' values
variables not in the Basis column	$= \mathbf{0}$
z row, RHS	current z (remember the units, and $z = -z'$ for a min)
final z row, under a slack/excess	**shadow price** (section 7)
final z row, under a non-basic decision variable	**reduced cost**
final z row, **0** under a **non-basic** variable	**alternative optima**

Basis = the list of variables in the Basis column, e.g. $\{x_1, x_2, s_3\}$.

Ratio test tie → pick either; the next table has a basic variable equal to 0 (degenerate). Carry on as normal.

5. Big M

1. **Test the origin** in every row. Rows that fail get an artificial.
2. Standard form: $+s$, $-e$, $+a$.
3. **Penalise:** max → $-\mathbf{Ma_i}$ in the objective, for **every** artificial. (Min: turn into max $z' = -z$ first, then $-Ma_i$.)
4. Write the z row: $z - c_1x_1 - \cdots + Ma_1 = 0$ (so $+M$ **under each artificial**).
5. **Clear the M's:** $R_z - M\times$ **(each artificial's row).**

Try	Entry under a	Result
$R_z + M \cdot R$	$M + M = 2M$	wrong
$R_z - M \cdot R$	$M - M = \mathbf{0}$	right

Reading it: a basic variable must have 0 in the z row, so **subtract**. Only the artificial rows are used, so only their RHS appears (e.g. $-12M$).

6. Run simplex. **Comparing M terms:** the bigger number in front of $-M$ is more negative ($-4 - 3M$ beats $-2 - M$). Unsure? Put $M = 1000$.
7. Once an artificial leaves, ignore its column.

End of Big M:

- artificial still basic with value > 0 → **infeasible**
- otherwise read the answer as normal (min: cost $= -z'$)

6. Special cases: the four tells

Situation	The tell in the table	What to write
Infeasible	stopped, but an **artificial is still basic with value** > 0	"no plan satisfies every constraint; they contradict each other"
Unbounded	entering column has **no positive entry**, so no ratio test	"profit can grow forever; the model is missing a limit"
Alternative optima	final z row has **0 under a non-basic variable**	"pivot it in to find a second best corner; every point between them is also optimal"
Degenerate	a basic variable $= 0$ (ratio tie)	"still fine; simplex may pivot without z changing"

"Already optimal" at the origin with an artificial basic = the M's were never cleared (Exam 2 Q5).

7. Sensitivity: shadow price, reduced cost, ranges

Where to look (final table)

Number under...	Name	Meaning
a slack s ($\leq$ constraint)	**shadow price**	z change per $+1$ of that RHS
an excess e ($\geq$ constraint)	shadow price, **sign flipped** for a max	see the table below
a non-basic decision variable x_j	**reduced cost**	how much z drops per unit if x_j is forced in
a **basic** slack (resource has spare)	0	shadow price **0**; more of it is worthless

Shadow price signs

Problem	Constraint	Look under	RHS $+1$ does
max	$\leq$ limit	s	profit **rises** by the number
max	$\geq$ requirement	e	profit **falls** by the number
min (solved as max $z' = -z$)	$\geq$ requirement	e	cost **rises** by the number
min (solved as max $z' = -z$)	$\leq$ limit	s	cost **falls** by the number
any	non-binding	–	**0**

Reading it: loosening a limit helps; tightening a requirement hurts. **Unsure? Re-solve with RHS $+1$: shadow price = new z − old z.**

Units: if z is in \$100s, a shadow price of 1 means **\$100**. Always convert.

RHS range (where the shadow price stays valid)

Change constraint i's RHS by Δ. Use **its slack column** (for a $\geq$, use **minus** its excess column):

$$\text{new basic value} = (\text{current RHS value}) + \Delta \times (\text{entry in that column}) \quad \geq \quad 0 \qquad \text{for every basic variable}$$

Furniture, timber, s_1 column $(2, -2, -3)$:

$$\begin{aligned} x_1 = 20 + 2\Delta \geq 0 \quad &\Rightarrow \quad \Delta \geq -10 \\ s_2 = 10 - 2\Delta \geq 0 \quad &\Rightarrow \quad \Delta \leq 5 \\ x_3 = 10 - 3\Delta \geq 0 \quad &\Rightarrow \quad \Delta \leq 10/3 \end{aligned}$$

$$\Rightarrow \; -10 \leq \Delta \leq \tfrac{10}{3} \; \Rightarrow \; \mathbf{40} \leq \text{timber} \leq \mathbf{53\tfrac{1}{3}} \qquad \text{(convert } \Delta \text{ back to the RHS!)}$$

- **Inside the range:** new z = old z + $\Delta\times$ shadow price. Say "inside the range" first.
- **Outside the range:** the formula is wrong. The basis changes, so **re-solve**.
- **"Buy k extra units at \$$P$ total?"** Check k is inside the range; gain = $k\times$ shadow price; **accept if gain** $> P$.

Reduced cost (a product not made)

$$\text{reduced cost} \quad = \quad (\text{its resources valued at shadow prices}) \; - \; (\text{its profit})$$

- Positive → losing money, don't make it. Its profit must rise **by the reduced cost** before it's worth making.
- **New product?** Same formula. **Negative → make it** (it earns more than its resources are worth elsewhere). This says *whether*, not *how many*: re-solve for that.
- Products you **do** make have reduced cost **0**.

Change	**Plan (x values)**	z
RHS (amount of a resource)	**moves** (same binding constraints, inside the range)	+ shadow price $\times\,\Delta$
profit of a product you make	**stays** (inside its range)	$+\,\Delta\times$ its amount
profit of a product you don't make	stays until the rise passes the reduced cost	unchanged

Excel Solver report

Column	**Meaning**
Final Value (variables)	the optimal plan
Reduced Cost	loss per unit if forced in (Solver shows it **negative** for a max)
Allowable Increase / Decrease (variables)	how far the profit coefficient can move with the **same plan**
Shadow Price	z change per +1 RHS
Allowable Increase / Decrease (constraints)	the RHS **range** where the shadow price holds

Sentences to write

- **Binding:** "Timber has shadow price \$100 per m^3: each extra m^3 adds \$100 of profit, valid between 40 and $53\frac{1}{3}$ m^3."
- **Non-binding:** "Labour has 10 days spare, so its shadow price is \$0; extra labour adds nothing."
- **$\geq$ requirement:** "Each extra kg the contract requires lowers profit by \$1."
- **Reduced cost:** "Tables have reduced cost \$300: forcing one in loses \$300, so a table's profit must rise from \$400 to at least \$700."
- **Which resource to buy:** the one with the **largest shadow price**, if it costs less than that.

8. Duality

Writing the dual (max with $\leq$ $\leftrightarrow$ min with $\geq$)

Primal (max)	**Dual (min)**
one constraint per resource	**one variable y_i per constraint** (value of that resource)
one variable per product	**one constraint per product**
RHS amounts	dual objective coefficients: $\min \mathbf{w} = \sum_{\mathbf{i}} \mathrm{RHS}_{\mathbf{i}}\, \mathbf{y_i}$
profits	dual RHS
a product's **column** (what it uses)	its dual constraint: **column** $\cdot\, y \geq$ **profit**
$\leq$, $x_j \geq 0$	$\geq$, $y_i \geq 0$
(rare) $\geq$ row / $=$ row in a max	$y_i \leq 0$ / y_i free

Reading it: **read DOWN the columns**, not across. Call the dual objective w. Write 0 coefficients explicitly.

Meaning to write: "y_i = the value of one unit of resource i, e.g. what a buyer would have to offer for it."

The theorems

Name	Statement	Use
Weak duality	any feasible $z \leq$ any feasible w	a dual-feasible y gives a **ceiling** on profit, without solving
Strong duality	optimal $z^* =$ optimal w^*	**certify optimal**: a feasible plan and a dual-feasible y with $z = w$ are **both** optimal
Dual values = shadow prices	$y^* =$ final z row under the slacks	read y^* straight off the tableau

Check a y is dual-feasible: plug it into every dual constraint ($\geq$ profit for each product) and $y \geq 0$. Then $w = \sum \text{RHS} \times y$ is an upper bound.

Complementary slackness

If...	then...	why
a resource has spare (slack > 0)	$\mathbf{y_i = 0}$	more of something you already have spare is worthless
$y_i > 0$	that constraint is **binding**	a resource only has value if you're short of it
a product is made ($x_j > 0$)	its dual constraint is **tight** ($=$)	made products break even at the shadow prices
a dual constraint has a **gap**	$\mathbf{x_j = 0}$, and gap = **reduced cost**	it loses the gap on every unit, so don't make it

Rule in one line: **(slack)** $\times y = 0$ and **(amount made)** $\times$ **(gap)** $= 0$.

Finding y^* from x^* (no tableau): furniture $x^* = (20, 0, 10)$, labour has spare.

labour spare	$\Rightarrow\ y_2 = 0$
chairs made (tight)	$\Rightarrow\ 2y_1 + 4y_2 + 3y_3 = 5\ \Rightarrow\ 2y_1 + 3y_3 = 5$
shelves made (tight)	$\Rightarrow\ y_1 + 2y_2 + 2y_3 = 3\ \Rightarrow\ y_1 + 2y_3 = 3$
solve	$\Rightarrow\ y_3 = 1,\ y_1 = 1\ \Rightarrow\ \mathbf{y^* = (1, 0, 1)},\quad w = 50 + 80 = 130 = z^*$ ✓

Why solve the dual? 400 products, 3 resources: primal = 400 variables, dual = **3 variables**. Much smaller, same optimal value, and it gives the shadow prices.

9. Transportation

1. **Balance:** total supply vs total demand.
   - supply > demand → **dummy destination**, cost 0 ("left at the source")
   - demand > supply → **dummy source**, cost 0 ("demand not met")
2. **Northwest corner:** start top-left, fill as much as possible, move right if the column is satisfied, down if the row is used up. Write the table and the **cost**.
3. **Check cells** $= m + n - 1$ (count the dummy). Too few (degenerate)? Put a 0 in a cell next to the corner where a row and column finished together.
4. **u, v:** set $u_1 = 0$; on **used** cells $\mathbf{u_i + v_j = c_{ij}}$.
5. **Reduced costs** on **empty** cells: $\mathbf{\bar{c}_{ij} = c_{ij} - u_i - v_j}$.
   - all $\bar{c} \geq 0 \rightarrow$ **optimal**; a $\bar{c} = 0 \rightarrow$ **alternative optimum**
   - else the **most negative** enters
6. **Loop:** from the entering cell, turn only at **used** cells, signs $+\ -\ +\ -$.
7. **θ = smallest amount on a $-$ cell.** Add θ to $+$, subtract from $-$. The cell hitting 0 leaves.
8. **New cost = old cost $+ \theta \times \bar{c}$.** Recheck the arithmetic by summing the new plan.

Draw the grid like this so you never flip back:

	S1 ($v = 7$)	S2 ($v = 8$)	Dummy ($v = 5$)
Q1 ($u = 0$)	\$7 → 40	\$4 [−4]	\$0 [−5]
Q2 ($u = -2$)	\$5 → 10	\$6 → 40	\$0 [−3]

Reading it: "\$cost → amount" for used cells, $[\bar{c}]$ for empty ones, u and v in the headers.

Why the loop works: each $+$ in a row/column is cancelled by a $-$ in the same row/column, so all totals stay right;

the cost change per unit = the $\bar{c}$.

10. Game theory

1. **Dominance first** (repeat until nothing changes):
   - **Rows** (row player wants **big**): row A $\geq$ row B in every column $\rightarrow$ delete B.
   - **Columns** (column player wants **small**): column A $\leq$ column B in every row $\rightarrow$ delete **B**.
2. **Saddle point test:**
   - **maximin** = biggest of the **row minimums**
   - **minimax** = smallest of the **column maximums**
   - equal $\rightarrow$ saddle point: the entry that is the **smallest in its row AND largest in its column**. Both play pure strategies; value = that number.
   - not equal $\rightarrow$ mixed strategy.
3. **Mixed, 2 rows:** $x_1 = P(\text{R1})$, $x_2 = 1 - x_1$. Set the payoffs against the two remaining columns **equal**, solve, then v = either side.
4. **Write the LP** (needed when asked, or for 3+ choices):

$$\begin{aligned} &\max\ v \\ &\text{s.t. (column 1 numbers)} \cdot x \geq v \qquad \leftarrow \text{one constraint per OPPONENT column} \\ &\quad \text{(column 2 numbers)} \cdot x \geq v \\ &\quad x_1 + \cdots + x_m = 1, \quad x \geq 0 \qquad (v \text{ any sign}) \end{aligned}$$

- Column player: **min** w, one constraint per **row**, row $\cdot\, y \leq w$. It's the **dual** of the row player's LP, so $v = w =$ value of the game.
- **Check:** the deleted column gives the row player $\geq v$ at your x.
- Answer with probabilities **and** the value **in words**: "R plays R1 4/9 of the time and R2 5/9; R gains 2/3 of a point per game on average."

11. Networks

Shortest path (label-setting / Dijkstra)

1. Box A = 0. Label its neighbours (distance, via).
2. Box the **unboxed node with the smallest total distance**.
3. From it, update each neighbour if (its distance + edge) is **smaller**. Mark rejected ones with $\times$.
4. Repeat until the destination is boxed. Trace back through "via".

Round	Box	Updates
1	A = 0	B 4 (A), C 3 (A)
2	C = 3	B 5 $\times$, D 11 (C), E 9 (C)
...	...	...

Reading it: one row per boxed node, so the marker can follow. Give the **route and its length**.

Minimum spanning tree (Prim)

Start anywhere; repeatedly add the **cheapest edge reaching a new node**; never close a loop. n nodes $\rightarrow n - 1$ **edges**. Give the edges with their lengths and the total.

	Shortest path	MST
Question	one route A $\rightarrow$ B	connect **all** nodes
Lock in each step	closest **node** (total distance)	cheapest **edge** to a new node

The LP behind shortest path

$x_{ij} = 1$ if edge $i \rightarrow j$ is used. $\min \sum \text{length} \times x_{ij}$, s.t. (out $-$ in) $= +1$ at the start, -1 at the end, 0 everywhere else (**conservation of flow**), $x \geq 0$.

12. Short-answer bank

Question	Answer
Why is the optimum at a corner?	The objective is **linear** (a flat plane, no interior peak). From any interior point you can move uphill until a constraint stops you, then along the edge until another stops you: a corner (or a whole edge if parallel).
Shadow price 0 means?	The constraint is **non-binding**: there's spare at the optimum, so one more unit can't help (complementary slackness).
0 in the z row under a non-basic **decision** variable?	**Alternative optima.** Pivot it in for a second optimal corner; report both and every point between.
Why solve the dual?	Fewer variables (the dual of a 400×3 problem has 3 variables); same optimal value (strong duality); gives shadow prices.
Why Big M / artificials?	$\geq$ and $=$ rows fail at the origin, so simplex has no starting corner. Artificials give a fake start; the $-M$ penalty forces them to 0.
Why must the M's be cleared?	Basic variables must have 0 in the z row; otherwise the z row is wrong and can falsely look optimal.
Why can a min have an optimum on an unbounded region?	With positive costs and $x \geq 0$, cost only grows as you move further out, so the minimum is reached at a corner of the region.
Why does a shadow price have a range?	Past the range a different constraint becomes binding, the corner changes, so the rate changes.
Why does the transportation starting plan need $m + n - 1$ cells?	One of the $m + n$ supply/demand equations is redundant (totals balance), so a basic solution has $m + n - 1$ basic cells.
What is a reduced cost, in words?	How much the objective worsens per unit if a product not being made is forced into the plan; equivalently, how much its profit must rise to be worth making.

13. Your personal traps

From marking your Practice Exam 2 (see `work/ans2_fixed.md`). **Check these before handing in.**

Trap	Fix
z in **\$100s**, wrote "\$1"	multiply: **1** $\rightarrow$ **\$100**, 130 $\rightarrow$ \$13,000
binding / non-binding backwards	**basic slack** > 0 $\rightarrow$ **spare**; **non-basic slack** $= 0$ $\rightarrow$ **fully used**
clearing M with +	$\mathbf{R_z} - \mathbf{M} \cdot$ (artificial row)
forgot $-M$ for one artificial	one $-Ma_i$ **per artificial**
only the final tableau	every tableau + entering/leaving + row ops
left out the "interpret / for the owner" part	finish every part with a sentence in words
"strong duality" with no numbers	read y^*, check feasibility, show $w = z$
checked x when asked about y	y is the **dual**: plug into **dual** constraints
saddle point from a matching number	must be **row min AND column max**
optimal set included infeasible points	limit the edge: e.g. $3 \leq x_1 \leq 6$
Δ range left as Δ	convert to the RHS range ($40 \leq$ timber $\leq 53\frac{1}{3}$) and say "inside the range"
transcribed a coefficient wrong	re-check every number against the question's table